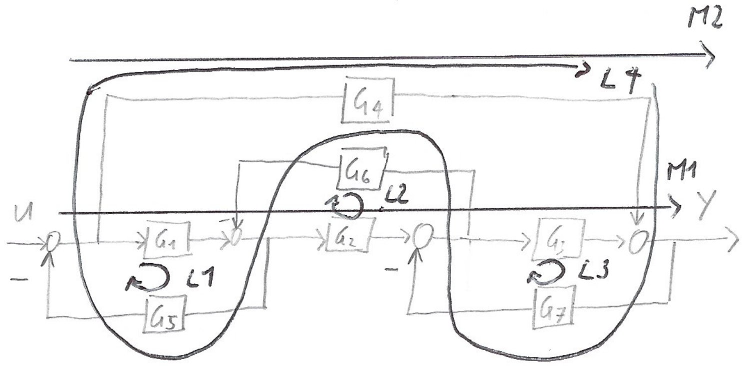
RS Vorlesung 2

HÜ 1)



$M1 = G_1 G_2 G_3$

$M2 = G_4$

$L1 = -G_1 G_5$

$L2 = -G_2 G_6$

$L3 = -G_3 G_7$

$L4 = -G_4 G_7 G_6 G_5$

$\Delta = 1 - (L1 + L2 + L3) + (L1 L3)$

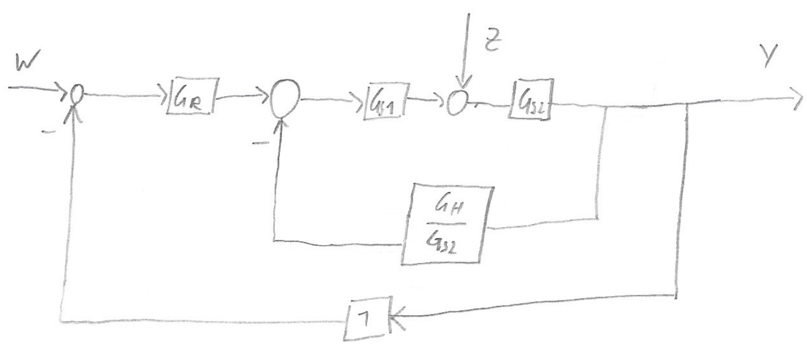
$H = \dfrac{\sum_j M_j \Delta_j}{\Delta}$

$\Delta_1 = 1$

$\Delta_2 = 1 - L2$

$$H = \frac{M1\,\Delta_1 + M2\,\Delta_2}{\Delta} = \frac{G_1 G_2 G_3 + G_4 + G_4 G_2 G_6}{1 + G_1 G_5 + G_2 G_6 + G_3 G_7 + G_1 G_3 G_5 G_7}$$

HÜ 2)



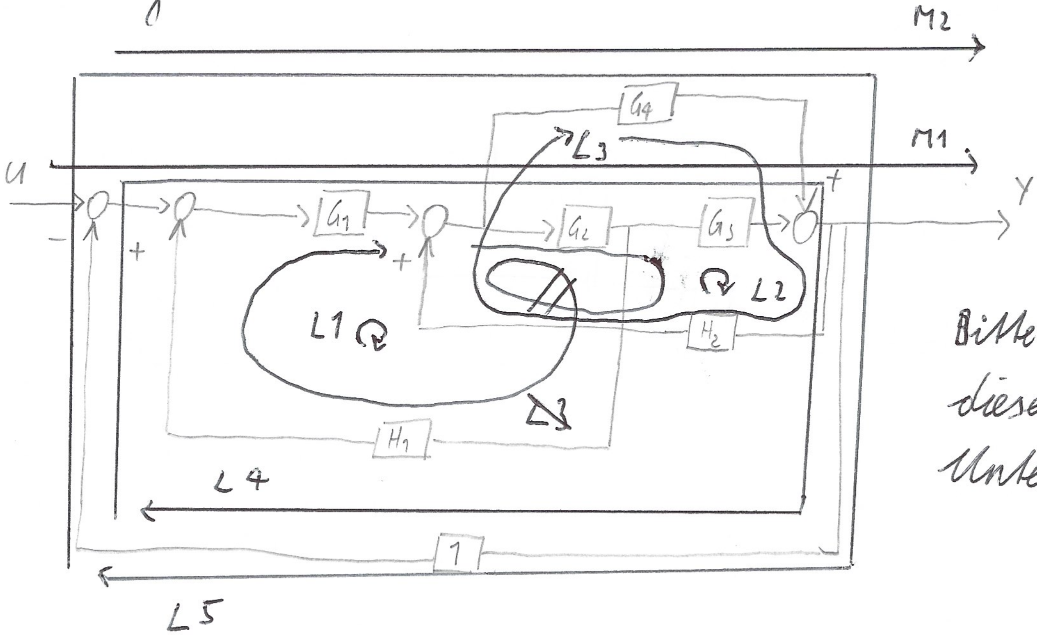
$$Y = G_{S2} \cdot \left(Z + G_{S1} \cdot \left(-Y \frac{G_H}{G_{S2}} + G_R \cdot (W - Y)\right)\right)$$

$$Y = G_{S2} Z - \cancel{G_{S2}} G_{S1} \frac{G_H}{\cancel{G_{S2}}} Y + G_{S2} G_R G_{S1} W - G_{S2} G_{S1} G_R Y$$

$$Y = \frac{G_{S2} Z + G_{S2} G_{S1} G_R W}{G_{S2} G_{S1} G_R + G_{S1} G_H + 1}$$

RS Vorlesung 2

HÜ 3)



Bitte lösen wir dieses Bsp. im Unterricht.

$M1 = G_1 G_2 G_3$ $\quad L_1 = G_1 G_2 H_1$ $\quad L_4 = -G_1 G_2 G_3$ $\quad L_3 = G_4 H_2$

$M2 = G_1 G_4$ $\quad L_2 = G_2 G_3 H_2$ $\quad L_5 = -G_1 G_4$

~~L_3~~ $= G_1$

$$\Delta = 1 - (L_1 + L_2 + L_3 + L_4 + L_5) + \cancel{(0)}$$

$\Delta_1 = 1$

$\Delta_2 = 1$

$$H = \frac{M1 + M2}{\Delta} = \frac{G_1 (G_2 G_3 + G_4)}{1 - G_1 G_2 H_1 - G_2 G_3 H_2 - G_4 H_2 + G_1 G_2 G_3 + G_1 G_4}$$